

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - II

Roll No. :S032	Name:Niraj Rane
Class :SYIT	Batch : 01
Date of Assignment :22/07/2026	Date/Time of Submission :22/07/2026

Questions:

2a. Write a python code to create nested tuples.

Code:

```
students = ("Niraj", 32, ("BscIT", "Semester III"), 90)

print("Name: ", students[0])
print("RollNo.: ", students[1])
print("Department: ", students[2][0])
print("Sem: ", students[2][1])
print("Percentage", students[-1])
```

Output:

```
===== RESTART: C:/Us
Name: Niraj
RollNo.: 32
Department: BscIT
Sem: Semester III
Percentage 90
|
```

2b. Write a python code to sort the nested tuple using sorted () function.

Code:

```
tup = ((1,3), (2,5), (6,2), (5,3))

Sorted_data = sorted(tup, key = lambda x: x[1])
print(Sorted_data)

print("NirajRane_s032")
```

Output:

```
===== RESTA
[(6, 2), (1, 3), (5, 3), (2, 5)]
NirajRane_s032
|
```

2c. Write a python code to copy or clone list.

Code:

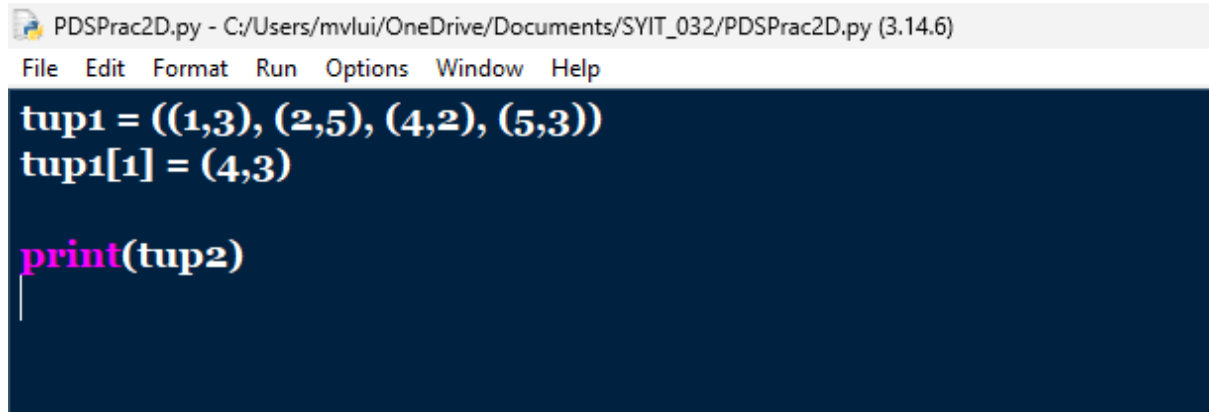
```
File Edit Format Run Options Window Help
tup1 = ((1,3), (2,5), (6,2), (5,3))
tup2 = ()
tup2 = tup1
print("Original List:", tup1)
print("Cloned List:", tup2)
```

Output:

```
===== RESTART
Original List: ((1, 3), (2, 5), (6, 2), (5, 3))
Cloned List: ((1, 3), (2, 5), (6, 2), (5, 3))
```

2d. Write a python code to check immutability property of python tuples.

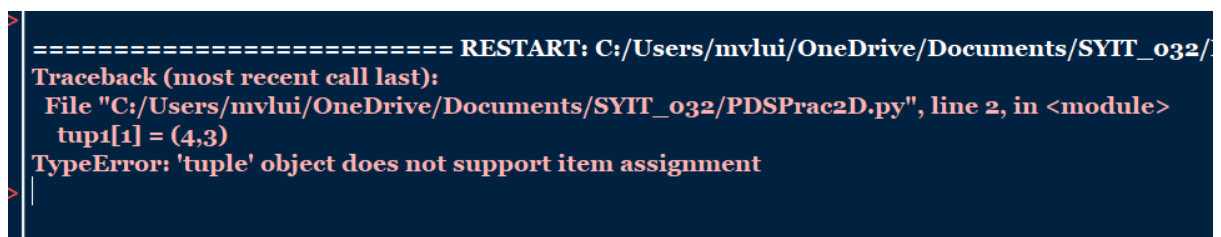
Code:

A screenshot of a Python IDE window titled 'PDSPrac2D.py - C:/Users/mvlui/OneDrive/Documents/SYIT_032/PDSPrac2D.py (3.14.6)'. The menu bar includes 'File', 'Edit', 'Format', 'Run', 'Options', 'Window', and 'Help'. The code editor has a dark blue background and contains the following Python code:

```
tup1 = ((1,3), (2,5), (4,2), (5,3))
tup1[1] = (4,3)

print(tup2)
```

Output:

A screenshot of a Python IDE window showing a runtime error. The window title is partially visible as 'PDSPrac2D.py - C:/Users/mvlui/OneDrive/Documents/SYIT_032/PDSPrac2D.py (3.14.6)'. The error message is displayed in a dark blue background with red text:

```
> ===== RESTART: C:/Users/mvlui/OneDrive/Documents/SYIT_032/PDSPrac2D.py (3.14.6)
Traceback (most recent call last):
  File "C:/Users/mvlui/OneDrive/Documents/SYIT_032/PDSPrac2D.py", line 2, in <module>
    tup1[1] = (4,3)
TypeError: 'tuple' object does not support item assignment
> |
```